Michael S. Truong

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Quantitative methods Ph.D. candidate (expected June 2027) who builds and validates statistical machine learning methods: tree-based structural equation modeling, large-scale Monte Carlo simulation on high-performance computing clusters, and simulation-based checks of LLM-generated statistical code. Seeking full-time data science, data engineering, or machine learning research roles in Canada. Canadian citizen based in Toronto, Ontario.

Education

Ph.D. in Psychology *(Quantitative Methods)* Expected June 2027
York University (Supervisor: *Dr. David B. Flora*)
Dissertation: *New Avenues in Machine Learning Structural Equation Modeling: IGSCA Trees*

M.A. in Psychology *(Developmental Sciences)* 2022
York University (Supervisors: *Drs. Thanujeni Pathman & W. Dale Stevens*)

H.B.Sc. in Neuroscience & Psychology 2020
University of Toronto (Supervisors: *Dr. Morris Moscovitch & Nichole Bouffard*)

Technical Skills

Programming: R (expert; package development, testthat), Stan, Quarto, Bash; Python and SQL (developing)
Statistics & Machine Learning: Tree-Based Machine Learning, Monte Carlo Simulation, Causal Inference, Bayesian Modeling, Structural Equation Modeling, Mixed-Effects Modeling, Categorical Data Analysis, Network Psychometrics
Data Engineering & HPC: Git, Singularity/Docker, Slurm (Digital Research Alliance of Canada), Arrow, DuckDB, targets pipelines, Linux
AI-Assisted Development: Claude Code; validating LLM-generated statistical code with Monte Carlo simulations

Professional & Research Experience

Doctoral Researcher 2022 - Present
York University (Supervisor: *Dr. David B. Flora*)

- Designed IGSCA Trees, a tree-based machine learning method that detects subgroups with different structural equation model parameters (dissertation; paper under review)
- Validated the method with large-scale Monte Carlo simulation studies, in collaboration with Drs. Florian Schuberth and Ji Yeh Choi
- Adapted simulation-based calibration to check the correctness of LLM-generated statistical software
- Compared frequentist and Bayesian multilevel models when cluster size is a confounder (paper with Dr. Xijuan (Cathy) Zhang), running three simulation studies that used 450+ CPU core-years on Digital Research Alliance of Canada clusters
- Presented at the International Meeting of the Psychometric Society (2025, 2026) and Modern Modeling Methods (2024)

Open-Source R Package Contributor 2023 - Present
github.com/emstruong

- cSEM R Package
  - Implemented IGSCA Trees and integrated GSCA-type models with the package's shared infrastructure
  - Reduced runtime of GSCA-type estimation routines by 50% while improving numerical stability
  - Updated testthat test infrastructure

- SimDesign R Package
  - Quality assurance on package features via GitHub bug reports (credited in package NEWS)
  - Filed several successful feature requests
  - Implementing improved handling of parallelized workloads via load balancing
- blavaan R Package
  - Facilitated reporting of model fits with new tidy *broom* methods
    - Initially drafted using Claude Opus and manually improved to match coding standards
  - Improved data analytical UX by adding `brms`-style default arguments
- lavaan R Package
  - Reported a bug in the handling of likelihood-ratio tests
  - Improved UX of fitting multigroup models
- glmmTMB R Package
  - Identified cross-platform differences in handling of model convergence

Statistical Consultant 2023 - 2026

York University, Statistical Consulting Service

- Advised graduate and faculty clients and gave weekly methods seminars, winning the Mirka Ondrack Award for contributions to statistical consulting

Peer-Review and Open Science 2022 - Present

- Reviewed manuscripts submitted to Q1 quantitative methods journals, including Advances in Methods and Practices in Psychological Science, for novelty, correctness, and fit
- Reviewed a public dataset for fidelity and worked with authors to obtain resolutions

Teaching Assistant and Guest Lecturer 2020 - Present

York University

- Practicum for Statistical Consulting (MATH6627); Univariate Analysis II: Regression (PSYC6132)

Research Assistant 2024 - 2025

York University (Supervisor: *Dr. Michael Friendly*)

- Reviewed *Visualizing Multivariate Linear Models*

Selected Publications

Exploring the Multiverse with Decision Trees Accepted, AMPPS

Truong, M. S., Crone, G., & Choi, J. Y.

Robust Recursive SEM with IGSCA Trees Under review

Truong, M. S., Schuberth, F., Choi, J. Y., & Flora, D. B.

What if Cluster Size is the Confound in Multilevel Models? In preparation

Truong, M. S., Zhang, X., & Flora, D. B.

Selected Awards

2025 SSHRC Doctoral Fellowship **$80,000**
2024 Mirka Ondrack Award for Contributions to Statistical Consulting
2023 Canadian Psychological Association: Quantitative Methods Student Presentation Award